

(\*)All the rings will be assumed to be commutative rings with identity if there is no special statement.

For a nonzero ring  $A$ , assume  $I_1 \subseteq I_2 \subseteq I_3 \subseteq \dots I_n \subseteq \dots$  is a chain of ideals in  $A$ , denote

$$\Sigma := \{I \subsetneq A \mid I \text{ is a proper ideal of } A\}$$

since  $(0) \in \Sigma$ ,  $\Sigma \neq \emptyset$ . In general, for a collection of ideals  $\{J_\lambda\}_{\lambda \in I}$ ,  $\bigcup_{\lambda \in I} J_\lambda$  may not be an ideal of  $A$ . However, for an increasing chain of ideals  $I_1 \subseteq I_2 \subseteq I_3 \subseteq \dots I_n \subseteq \dots$ , we can check that  $\bigcup_{n=1}^{\infty} I_n$  is an ideal of  $A$ . Furthermore, if  $I_n$  is a proper ideal of  $A$  for  $\forall n \in \mathbb{N}$ , then  $\bigcup_{n=1}^{\infty} I_n$  is a proper ideal of  $A$  (if  $\bigcup_{n=1}^{\infty} I_n = A$ , then  $1 \in \bigcup_{n=1}^{\infty} I_n$ , hence  $1 \in I_m$  for some  $m \in \mathbb{N}$ , which implies that  $I_m = A$ , a contradiction!). By Zorn's Lemma, there exists a maximal element of  $\Sigma$ , denoted by  $\mathfrak{m}$ , thus  $\mathfrak{m}$  is a maximal ideal of  $A$ .

Hence we get the theorem

**Theorem 0.1.** : *If  $A$  is a nonzero ring, then it has a maximal ideal.*

**Corollary 0.1.** : *If  $A$  is a nonzero ring,  $I \subsetneq A$  is a proper ideal of  $A$ , then there exists a maximal ideal of  $A$  containing  $I$ .*

**Corollary 0.2.** : *If  $x \in A$  is not a unit, then there exists a maximal ideal of  $A$  containing  $x$ .*

Proof: Apply the corollary 0.1 to the ideal  $(x)$ , the ideal generated by  $x$ .  $\square$

Let  $I \triangleq \{x \in A \mid x \text{ is not a unit in } A\}$ . In the case that  $I$  is an ideal of  $A$ , then  $I$  must be a proper ideal, for  $x \in I$ , by Corollary 0.1, there exists a maximal ideal of  $A$  containing  $x$ , say  $\mathfrak{m}$ . For  $\forall y \in \mathfrak{m}$ , since  $y$  is not a unit (because  $\mathfrak{m}$  is a proper ideal), hence  $y \in I$ , thus  $\mathfrak{m} \subseteq I$ , which means  $I = \mathfrak{m}$ . Also,  $I$  is the only maximal ideal of  $A$ .

Hence we get the proposition.

**Proposition 0.1.** : *If the set of all non-units, denoted by  $I$ , forms an ideal of  $A$ , then  $I$  is the unique maximal ideal of  $A$ .*

## Local Ring

**Definition 0.1.** : A ring is said to be a **local ring** if it has only one maximal ideal.

**Proposition 0.2.** : Let  $A$  be a ring,  $\mathfrak{m}$  is a maximal ideal in  $A$ , if  $1 + x$  is a unit in  $A$  for  $\forall x \in \mathfrak{m}$ , then  $\mathfrak{m}$  is the unique maximal ideal in  $A$ .

Proof: let  $x$  be a non-unit, if  $x \notin \mathfrak{m}$ , then  $\mathfrak{m} + (x) = A$ , thus there exist  $u \in \mathfrak{m}$  and  $r \in A$  such that  $u + rx = 1$ , hence  $1 - u = rx$ , which implies by the condition that  $1 - u$  is a unit, hence  $rx$  is a unit,  $x$  is a unit (contradiction to our assumption!), so we must have  $x \in \mathfrak{m}$ . Therefore, we get  $\mathfrak{m} = \{x \in A | x \text{ is not a unit}\}$ . By proposition 0.1,  $\mathfrak{m}$  is the unique maximal ideal of  $A$ , hence  $A$  is a local ring.  $\square$

Let  $\mathfrak{n} := \{x \in A | x^n = 0 \text{ for some } n \in \mathbb{N}\}$ , for  $\forall x, y \in \mathfrak{n}$ , there exist  $n, m \in \mathbb{N}$  such that  $x^n = 0, y^m = 0$ , hence we have  $(x + y)^{m+n} = 0$ , which means  $x + y \in \mathfrak{n}$ ; for  $\forall x \in \mathfrak{n}$ , it is clear that  $-x \in \mathfrak{n}$ ; for  $\forall x \in \mathfrak{n}$  and  $r \in A$ , then it's easy to see  $rx \in \mathfrak{n}$ .

Hence we get the proposition

**Proposition 0.3.** : The set of all nilpotents, denoted by  $\mathfrak{n}$ , forms a proper ideal of  $A$ .

Suppose  $P$  is a prime ideal of  $A$ , if  $x \in \mathfrak{n}$ , then  $x^n = 0$  for some  $n \in \mathbb{N}$ , since  $x^n = 0 \in P$ , we can know that  $x \in P$ , which implies that

$$\mathfrak{n} \subseteq \bigcap_{P \text{ is prime ideal}} P;$$

Now suppose  $x \notin \mathfrak{n}$ , then  $x^n \neq 0$  for  $\forall n \in \mathbb{N}$ . Consider the set

$$\Sigma = \{I \subsetneq A | I \text{ is a proper ideal of } A\},$$

since  $(0) \in \Sigma$ ,  $\Sigma \neq \emptyset$ , let  $I_1 \subseteq I_2 \subseteq I_3 \subseteq \dots \subseteq I_n \subseteq \dots$  be a chain in  $\Sigma$ , then  $I \triangleq \bigcup_{n=1}^{\infty} I_n$  is a proper ideal of  $A$  and an upper bound of  $\Sigma$ , by Zorn's Lemma,  $\Sigma$  has a maximal ideal, denoted by  $P$ . Let  $ab \in P$ , if  $a \notin P$  and  $b \notin P$ , then  $P + (a) \notin \Sigma$ ,  $P + (b) \notin \Sigma$ , thus there exist  $n, m \in \mathbb{N}$  such that  $x^n \in P + (a)$  and  $x^m \in P + (b)$ , hence  $x^n = z_1 + r_1a$  and  $x^m = z_2 + r_2b$  for some  $z_1, z_2 \in P$  and  $r_1, r_2 \in A$ , then  $x^{n+m} = z_1z_2 + z_1r_2b + z_2r_1a + r_1r_2ab \in P$  (since  $P$  is an ideal), this is a contradiction! Hence for  $ab \in P$ , we have  $a \in P$  or  $b \in P$ , which means  $P$  is a prime ideal and  $x \notin P$ .

By the discussion above, we know that

$$x \notin \mathfrak{n} \Rightarrow x \notin \bigcap_{P \text{ is a prime ideal}} P,$$

thus

$$\bigcap_{P \text{ is a prime ideal}} P \subseteq \mathfrak{n}$$

Hence we get the proposition

**Proposition 0.4.** :Let  $\mathfrak{n}$  be the set of all nilpotents, then

$$\mathfrak{n} = \bigcap_{P \text{ is a prime ideal}} P$$

### Jacobson radical

**Definition 0.2.** :For a ring  $A$ ,

$$J(A) := \bigcap_{\mathfrak{m} \text{ is a maximal ideal}} \mathfrak{m}$$

is called the **Jacobson radical** of  $A$ .

Let  $x \in J(A)$ , then  $x \in \mathfrak{m}$  ( $\forall \mathfrak{m}$  maximal ideal). For  $\forall y \in A$ , we must have  $1 - xy \notin \mathfrak{m}$  (or otherwise,  $1 - xy \in \mathfrak{m}$  will imply  $1 \in \mathfrak{m}$ , which is a contradiction!), hence  $1 - xy$  is a unit of  $A$  (since every non-unit is contained in some maximal ideal of  $A$ );

On the other hand, if for  $y \in A$ ,  $1 - xy$  is a unit in  $A$ , suppose  $x \notin \mathfrak{m}$  for some maximal ideal  $\mathfrak{m}$  of  $A$ , then  $\mathfrak{m} + (x) = A$ , i.e., there exists  $u \in \mathfrak{m}$  and  $r \in A$  such that  $u + rx = 1$ , which means  $u = 1 - rx$  is a unit. (contradiction!), which implies that  $x \in J(A)$ .

Hence we get the proposition

**Proposition 0.5.** :For a ring  $A$ ,  $J(A) = \{x \in A \mid 1 - xy \text{ is a unit of } A \text{ for } \forall y \in A\}$ .

**Corollary 0.3.** :Let  $\mathfrak{m}$  be maximal ideal and  $1 + x$  is a unit for  $\forall x \in \mathfrak{m}$ , then  $\mathfrak{m} = J(A)$ , i.e.,  $\mathfrak{m}$  is the unique maximal ideal of  $A$  and  $A$  is a local ring.

Proof: If  $1 + x$  is a unit for  $\forall x \in \mathfrak{m}$ , let  $y$  be a non-unit of in  $A$ . Then  $xy \in \mathfrak{m}$  and  $1 + xy$  must be a unit. By the proposition above,  $x \in J(A)$ . Thus  $\mathfrak{m} \subseteq J(A)$ , which means that  $J(A) = \mathfrak{m}$  (by the definition of  $J(A)$ ) and  $A$  is a local ring.  $\square$

We talk about the operations on ideals:

### Operations on Ideals

For a ring  $A$ , let  $I, J$  be two ideals of  $A$ .

(1) for the two ideals  $I$  and  $J$ ,  $I \cup J$  need not to be an ideal of  $A$ ;  $I \cup J$  is still an ideal  $\Leftrightarrow I \subseteq J$  or  $J \subseteq I$ .

(2)  $I + J = \{x + y | x \in I, y \in J\}$  is an ideal of  $A$ .

(3)  $I \cap J = \{x \in A | x \in I \text{ and } x \in J\}$  is an ideal of  $A$ .

(4)  $IJ :=$  smallest ideal containing the set  $\{xy | x \in I, y \in J\} = \{\sum_i x_i y_i (\text{finite sum}) | x_i \in I, y_i \in J\}$ .

**Example 0.1.**  $m\mathbb{Z} + n\mathbb{Z} = \gcd(m, n)\mathbb{Z}$ ,  $m\mathbb{Z} \cap n\mathbb{Z} = \text{lcm}(m, n)\mathbb{Z}$ .

**Remark 0.1.** In general, we do not have  $(I + J)(I \cap J) = IJ$ , but when  $I + J = A$ , the equality holds.

*Proof:* Suppose  $I + J = A$ , then  $(I + J)(I \cap J) = A(I \cap J) = I \cap J \supseteq IJ$ , on the other hand, for  $\forall a_i \in A = I + J$ ,  $\exists x_i \in I$  and  $y_i \in J$  such that  $a_i = x_i + y_i$ . Then for  $\forall r_i \in I \cap J$ ,  $a_i r_i = (x_i + y_i)r_i$ , which means that  $(I + J)(I \cap J) = A(I \cap J) = I \cap J \subseteq IJ$ .  $\square$

### Coprime Ideals

**Definition 0.3.** For a ring  $A$ ,  $I, J$  are ideals of  $A$ , if  $I + J = A$ , then  $I$  and  $J$  are said to be **coprime**.

For a ring  $A$ , let  $I, J$  be two ideals of  $A$ . Consider the homomorphism

$$\varphi : A \rightarrow A/I \times A/J$$

$$x \mapsto (x + I, x + J)$$

then  $\ker \varphi = I \cap J$ . Assume  $I$  and  $J$  are coprime, i.e.,  $I + J = A$ . For  $\forall (a + I, b + J) \in A/I \times A/J$ , since  $I + J = A$ ,  $\exists x \in I$  and  $\exists y \in J$  such that  $x + y = 1$ . Then  $bx = b - by$ ,  $ay = a - ax$ ,  $bx + ay = (b - by) + (a - ax) = a - ax + b(1 - y) \in a + I$ . Similarly,  $bx + ay = b - by + a(1 - x) \in b + J$ , which implies that  $\varphi(bx + ay) = (a + I, b + J)$ ,  $\varphi$  is an epimorphism.

By the discussion above, we have the proposition

**Proposition 0.6.** For a ring  $A$ , let  $I_1, I_2$  be two ideals of  $A$ . Consider the homomorphism

$$\varphi : A \rightarrow A/I_1 \times A/I_2$$

$$x \mapsto (x + I_1, x + I_2)$$

then we have

(1) if  $I_1$  and  $I_2$  are coprime, i.e.,  $I_1 + I_2 = A$ , then  $I_1 \cap I_2 = I_1 I_2$ ;

(2)  $\varphi$  is surjective  $\Leftrightarrow I_1 + I_2 = A$ , i.e., they are coprime;

(3)  $\varphi$  is injective  $\Leftrightarrow \ker \varphi = I_1 \cap I_2 = \{0\}$ .

By induction, similarly, we have the following proposition

**Proposition 0.7.** :Let  $I_1, I_2, \dots, I_n$  be ideals of  $A$ , consider the homomorphism

$$\varphi : A \rightarrow A/I_1 \times A/I_2 \times \dots \times A/I_n$$

$$x \mapsto (x + I_1, x + I_2, \dots, x + I_n)$$

then we have

- (1) if  $I_i$  and  $I_j$  are coprime for  $\forall i \neq j$ , i.e.,  $I_i + I_j = A$ , then  $\prod_{i=1}^n I_i = \cap_{i=1}^n I_i$ ;
- (2)  $\varphi$  is surjective  $\Leftrightarrow I_i$  and  $I_j$  are coprime for  $\forall i \neq j$ ;
- (3)  $\varphi$  is injective  $\Leftrightarrow \ker \varphi = \cap_{i=1}^n I_i = \{0\}$ .

Let  $I$  be an ideal of a ring  $A$ ,  $P_1$  and  $P_2$  are two prime ideals of  $A$  such that  $I \subseteq P_1 \cup P_2$ . Take arbitrary  $x \in I$ , assume that  $I \not\subseteq P_1$ , which means that we can take an element  $y \in I$  and  $y \notin P_1$ , since  $I \subseteq P_1 \cup P_2$ ,  $y \in P_2$ . It's clear that  $x + y \in I \subseteq P_1 \cup P_2$ . Suppose  $x + y \in P_1$ , if  $x \in P_1$ , then  $(x + y) - x = y \in P_1$  (contradiction!), therefore  $x \notin P_1$  and  $x \in P_2$ , by the arbitrariness of  $x \in I$ , we get  $I \subseteq P_2$ ; suppose  $x + y \notin P_1$ , then  $x + y \in P_2$ , therefore  $(x + y) - y = x \in P_2$ , thus  $I \subseteq P_2$ .

Hence we get the proposition

**Proposition 0.8.** :Let  $P_1$  and  $P_2$  be two prime ideals of a ring  $A$ ,  $I$  is an ideal of  $A$  such that  $I \subseteq P_1 \cup P_2$ , then we have  $I \subseteq P_1$  or  $I \subseteq P_2$ .

**Proposition 0.9.** :Let  $P_1, P_2, \dots, P_n$  be prime ideals of a ring  $A$ ,  $I$  is an ideal such that  $I \subseteq P_1 \cup P_2 \cup \dots \cup P_n$ , then  $\exists 1 \leq j \leq n$  such that  $I \subseteq P_j$ .

Proof: By induction.

When  $n = 2$ , the conclusion has been proved in the discussion above. Now suppose the conclusion holds for the case that  $n - 1$ . In the case of  $n$ , if  $I \subseteq P_{i_1} \cup P_{i_2} \cup \dots \cup P_{i_{n-1}}$  for some index set  $\{i_1, i_2, \dots, i_{n-1}\} \subseteq \{1, 2, \dots, n\}$ , then by the induction hypothesis,  $I \subseteq P_{i_j}$  for some  $i_j \in \{i_1, i_2, \dots, i_{n-1}\}$ . If  $I \not\subseteq P_{i_1} \cup P_{i_2} \cup \dots \cup P_{i_{n-1}}$  for any index set  $\{i_1, i_2, \dots, i_{n-1}\} \subseteq \{1, 2, \dots, n\}$ , then take the elements  $x_i \in I \setminus \bigcup_{j=1, j \neq i}^n P_j$  ( $i = 1, 2, \dots, n$ ), i.e.,  $x_1 \notin P_j (\forall j \neq 1), x_2 \notin P_j (\forall j \neq 2), \dots$ . Without loss of generality, we may take the case that  $I \not\subseteq P_1 \cup P_2 \cup \dots \cup P_{n-1}$ , it is clear that  $x_1 + x_2 + \dots + x_n \in I$ , thus  $\exists 1 \leq j \leq n$  such that  $x_1 + x_2 + \dots + x_n \in P_j$ . Consider the product element  $x_1 x_2 \dots x_{n-1}$ , it's obvious that  $x_1 x_2 \dots x_{n-1} \in P_s$  for  $\forall 1 \leq s \leq n-1$  and  $x_1 x_2 \dots x_{n-1} \notin P_n$  (otherwise, if  $x_1 x_2 \dots x_{n-1} \in P_n$ , then there must exists some  $x_t \in P_n$  with  $1 \leq t \leq n-1$  since  $P_n$  is a prime ideal). Consider the element  $y := x_2 x_3 \dots x_n + x_1 x_3 \dots x_n + \dots + x_1 x_2 \dots x_{n-1} \in I \subseteq P_1 \cup P_2 \cup \dots \cup P_n$ , hence  $\exists 1 \leq j \leq n$  such that  $y \in P_j$ , then we must have  $x_1 x_2 \dots x_{j-1} x_{j+1} \dots x_n \in P_j$  since  $x_1 x_2 \dots x_{j-1} x_{j+1} \dots x_n = y - x_2 x_3 \dots x_n - \dots - x_1 x_2 \dots x_{n-1}$  (note that the other product

elements are all in  $P_j$ ), which is a contradiction! Therefore, the discussion implies that there must exist some index set  $\{i_1, i_2, \dots, i_{n-1}\} \subseteq \{1, 2, \dots, n\}$  such that  $I \subseteq P_{i_1} \cup P_{i_2} \cup \dots \cup P_{i_{n-1}}$ . Thus by the induction hypothesis,  $\exists k$  such that  $I \subseteq P_k$ .  $\square$

**Proposition 0.10.** : Let  $P$  be a prime ideal,  $I_1, I_2$  are two ideals such that  $I_1 \cap I_2 \subseteq P$ , then we have  $I_1 \subseteq P$  or  $I_2 \subseteq P$ . Furthermore, if  $I_1 \cap I_2 = P$ , then we have  $I_1 = P$  or  $I_2 = P$ .

Proof: Let  $P$  be a prime ideal of the ring  $A$ ,  $I_1, I_2$  are two ideals such that  $I_1 \cap I_2 \subseteq P$ , assume that  $I_1 \not\subseteq P$ , take  $\forall y \in I_2$ , there exists some element  $x \in I_1 \setminus P$ , then  $xy \in I_1 \cap I_2 \subseteq P$ , hence  $y \in P$  since  $x \notin P$ , which means  $I_2 \subseteq P$ ;

If  $I_1 \cap I_2 = P$ , then  $I_1 \cap I_2 \supseteq P$ ,  $P \subseteq I_1$  and  $P \subseteq I_2$ , hence  $I_1 = P$  or  $I_2 = P$ .  $\square$

**Proposition 0.11.** : Let  $P$  be a prime ideal of the ring  $A$ ,  $I_1, I_2, \dots, I_n$  are ideals of  $A$  such that  $\bigcap_{i=1}^n I_i \subseteq P$ , then we have  $I_i \subseteq P$  for some  $1 \leq i \leq n$ . Furthermore, if  $\bigcap_{i=1}^n I_i = P$ , then we have  $I_i = P$  for some  $1 \leq i \leq n$ .

Proof: By induction.  $\square$